



AS Level

Paper 1 - Solutions

9709/01



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QUADRATICS

Completing the square

9709_w21_13 -q3

Express $5y^2 - 30y + 50$ in the form $5(y + a)^2 + b$, where a and b are constants.

$$\begin{aligned} & 5y^2 - 30y + 50 \\ & \underline{5(y^2 - 6y + \frac{6}{2}^2) + (5 \times -(3)^2 + 50)} \\ & \underline{5(y^2 - 6y + 3^2) + (-45 + 50)} \\ & \underline{5(y - 3)^2 + 5} \\ & \underline{a = -3} \\ & \underline{b = 5} \end{aligned}$$

9709_s18_13 -q1

Express $3x^2 - 12x + 7$ in the form $a(x + b)^2 + c$, where a , b and c are constants.

$$\begin{aligned} & 3x^2 - 12x + 7 \\ & \underline{3(x^2 - 4x + \frac{4}{2}^2) + (3 \times -(2)^2 + 7)} \\ & \underline{3(x^2 - 4x + 2^2) + (-12 + 7)} \\ & \underline{3(x - 2)^2 - 5} \\ & \underline{a = 3 \quad c = -5} \\ & \underline{b = -2} \end{aligned}$$

Discriminant:

9709_S14_11

Find the set of values of k for which the line $y = 2x - k$ meets the curve $y = x^2 + kx - 2$ at two distinct points. [5]

① Equate both equations

$$2x - k = x^2 + kx - 2$$

② Form a quadratic

$$x^2 - 2x + kx - 2 + k = 0$$

$$x^2 + (k-2)x - 2 + k = 0$$

③ Use discriminant " $b^2 - 4ac > 0$ "

$$(k-2)^2 - 4(1)(-2+k) > 0$$

$$k^2 - 4k + 4 + 8 - 4k > 0$$

$$k^2 - 8k + 12 > 0$$

$$(k-6)(k-2) > 0$$

$$k > 6 \quad k > 2$$

flip inequality of smaller number

$$k > 6 \text{ and } k < 2$$

Q4) 9709_M17_12

Find the set of values of k for which the equation $2x^2 + 3kx + k = 0$ has distinct real roots.

$$2x^2 + 3kx + k = 0$$

$$b^2 - 4ac \geq 0$$

$$(3k)^2 - 4(2)(k) \geq 0$$

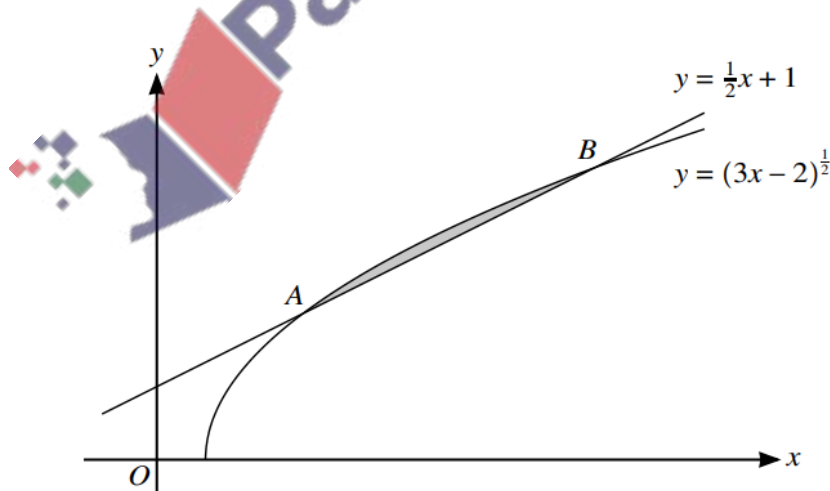
$$9k^2 - 8k \geq 0$$

$$K(9K - 8) > 0$$

flip! $\hookrightarrow \begin{matrix} K > 0 & K \approx 0 \\ K < 0 & \end{matrix}$

Solutions To A Graph:

Q5) 9709_s22_qp_11



The diagram shows the curve with equation $y = (3x - 2)^{\frac{1}{2}}$ and the line $y = \frac{1}{2}x + 1$. The curve and the line intersect at points A and B .

(a) Find the coordinates of A and B .

[4]

• Equate both equations

$$(3x-2)^{1/2} = 0.5x + 1$$

$$(3x-2) = (0.5x+1)^2$$

$$(3x-2) = (0.25x^2 + x + 1)$$

$$0.25x^2 - 3x + x + 2 + 1$$

$$\frac{1}{4}x^2 - 2x + 3 = 0$$

$$x = 6 \quad \swarrow \text{middle term} \quad \searrow \quad x = 2$$

$$y = \frac{1}{2}(6) + 1 = 4$$

$$y = \frac{1}{2}(2) + 1 = 2$$

$$(6, 4)$$

$$(2, 2)$$

Q6) 9709_w21_qp_13

The line $y = 2x + 5$ intersects the circle with equation $x^2 + y^2 = 20$ at A and B.

(a) Find the coordinates of A and B in surd form and hence find the exact length of the chord AB.

[7]

LINE AND CIRCLE

① EQUATE

$$y = 2x + 5 \quad y = \sqrt{20 - x^2}$$

$$2x + 5 = (20 - x^2)^{1/2}$$

$$(2x + 5)^2 = 20 - x^2$$

$$4x^2 + 20x + 25 = 20 - x^2$$

$$5x^2 + 20x + 5 = 0$$

$$x = -2 + \sqrt{3}$$

$$x = -2 - \sqrt{3}$$

$$y = 2(-2 + \sqrt{3}) + 5 = 1 + 2\sqrt{3}$$

$$y = 2(-2 - \sqrt{3}) + 5 = 1 - 2\sqrt{3}$$

$$A (-2 + \sqrt{3}, 1 + 2\sqrt{3})$$

$$B (-2 - \sqrt{3}, 1 - 2\sqrt{3})$$

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = 6$$

COORDINATE GEOMETRY

Equation Of Perpendicular Bisector Of A Line

Q1) 9709_S19_QP_12

Two points A and B have coordinates $(1, 3)$ and $(9, -1)$ respectively. The perpendicular bisector of AB intersects the y -axis at the point C . Find the coordinates of C . [5]

$$8) A(1, 3) \quad B(9, -1)$$

$$m = \frac{-1-3}{9-1} = -\frac{1}{2}$$

① make equation of perpendicular bisector using gradient and midpoint.

gradient $\rightarrow 2$

$$\text{midpoint} \rightarrow \left(\frac{1+9}{2}, \frac{3+(-1)}{2} \right) = (5, 1)$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 2(x - 5)$$

$$y = 2x - 9$$

② Put ' $x=0$ ' to find y -axis intersection.

$$y = 2x - 9$$

$$y = 2(0) - 9$$

$$y = -9$$

$$C = (0, -9)$$

Using Expanded Formula Of Circle

Q2) 9709_s22_qp_11

The equation of a circle is $x^2 + y^2 + 6x - 2y - 26 = 0$.

- (a) Find the coordinates of the centre of the circle and the radius. Hence find the coordinates of the lowest point on the circle. [4]

$$\begin{aligned} x^2 + y^2 + 6x - 2y - 26 &= 0 \\ \text{(i) arrange} \quad x^2 + 6x + y^2 - 2y &= 26 \\ \text{(ii) form identity.} \quad x^2 + 6x + \left(\frac{6}{2}\right)^2 + y^2 - 2y + \left(\frac{2}{2}\right)^2 &= 26 + \left(\frac{6}{2}\right)^2 + \left(\frac{2}{2}\right)^2 \\ \text{(iii) solve} \quad x^2 + 6x + 3^2 + y^2 - 2y + 1 &= 26 + 9 + 1 \\ \text{(iv) find centre \& radius} \quad (x + 3)^2 + (y - 1)^2 &= 36 \\ \text{centre} &\rightarrow (-3, 1) \\ \text{radius} &\rightarrow \sqrt{36} = 6 \end{aligned}$$

Circle Properties

Q3) 9709_s20_qp_11

The coordinates of the points A and B are $(-1, -2)$ and $(7, 4)$ respectively.

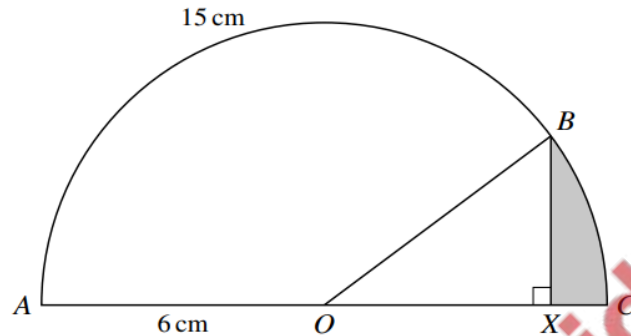
- (a) Find the equation of the circle, C, for which AB is a diameter. [4]

$$\begin{aligned} A(-1, -2) \quad B(7, 4) \\ 1. \text{ Centre is the midpoint of these 2 points.} \\ \text{m.p} &\rightarrow \left(\frac{7 + (-1)}{2}, \frac{4 + (-2)}{2} \right) \\ &= 3, 1 \\ 2. \text{ radius is distance from midpoint to point on circle} \\ &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(7 - 3)^2 + (1 - 4)^2} \\ &= \sqrt{16 + 9} \\ &= \sqrt{25} \\ &= 5 \\ 3. \text{ substitute values in equation form.} \\ (x - x_1)^2 + (y - y_1)^2 &= r^2 \\ (x - 3)^2 + (y - 1)^2 &= 25 \quad \text{[Ans]} \end{aligned}$$

CIRCULAR MEASURE

Finding Arc Length of Shaded Region

Q) 9709_s20_qp_11



In the diagram, ABC is a semicircle with diameter AC , centre O and radius 6 cm. The length of the arc AB is 15 cm. The point X lies on AC and BX is perpendicular to AX .

Find the perimeter of the shaded region BXC .

[6]

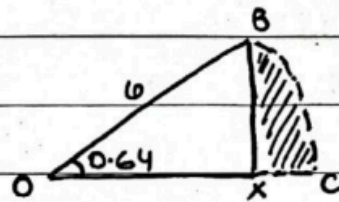
i) Find $\angle BOC$ using $s = r\theta$

$$s = r\theta$$

$$15 = 6\theta$$

$$\theta = 2.5$$

$$\pi - \theta = 0.64 = \angle BOC$$



ii) $BX \rightarrow 6 \sin(0.64) = 3.58$

$OX \rightarrow 6 \cos(0.64) = 4.80$

$XC \rightarrow 6 - 4.80 = 1.19$

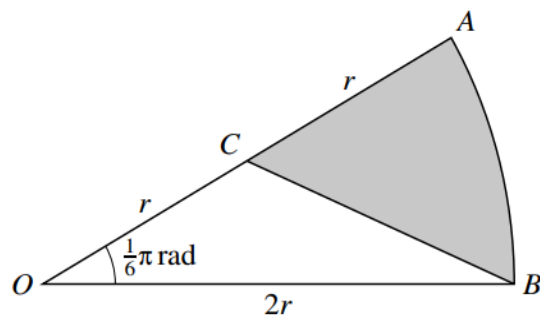
$BC \rightarrow r\theta = 6(0.64) = 3.85$

iii) Perimeter : $BX + XC + BC$

$$3.58 + 1.19 + 3.85$$

$$8.634$$

Q) 9709_s20_qp_12



In the diagram, OAB is a sector of a circle with centre O and radius $2r$, and angle $AOB = \frac{1}{6}\pi$ radians. The point C is the midpoint of OA .

(a) Show that the exact length of BC is $r\sqrt{5 - 2\sqrt{3}}$.

[2]

1. Use cosine rule:

$$c^2 = a^2 + b^2 - 2ab(\cos \theta)$$

$$BC^2 = r^2 + (2r)^2 - 2(r)(2r)(\cos \pi/6)$$

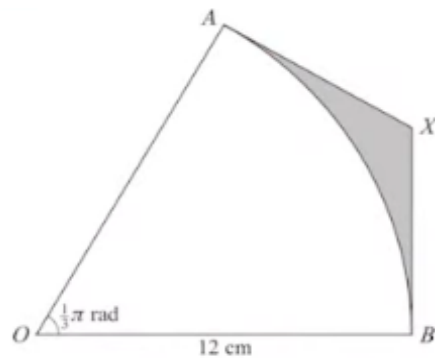
$$BC^2 = 5r^2 - 2\sqrt{3}r^2$$

$$BC^2 = r^2(5 - 2\sqrt{3})$$

$$BC = \sqrt{r^2(5 - 2\sqrt{3})}$$

$$BC = r\sqrt{5 - 2\sqrt{3}}$$

FINDING AREA OF SHADED REGION :



In the diagram, OAB is a sector of a circle with centre O and radius 12 cm . The lines AX and BX are tangents to the circle at A and B respectively. Angle $AOB = \frac{1}{3}\pi$ radians.

- (i) Find the exact length of AX , giving your answer in terms of $\sqrt{3}$. [2]
- (ii) Find the area of the shaded region, giving your answer in terms of π and $\sqrt{3}$. [3]

Solution:

(i) $AX \rightarrow \tan \theta = \frac{\text{Opp}}{\text{Adj}}$

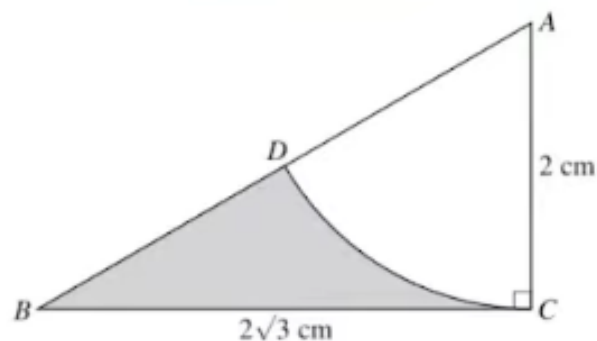
$$\tan\left(\frac{\pi}{6}\right) = \frac{AX}{12}$$

$$4\sqrt{3} = AX$$

(ii) $(2 \times \text{area of } \Delta) - (\text{area of sector})$

$$\left(2 \times \frac{1}{2} \times 12 \times 4\sqrt{3} \right) - \left(\frac{1}{2} \times 12^2 \times \frac{\pi}{3} \right)$$

$$48\sqrt{3} - 24\pi$$



In the diagram, D lies on the side AB of triangle ABC and CD is an arc of a circle with centre A and radius 2 cm. The line BC is of length $2\sqrt{3}$ cm and is perpendicular to AC . Find the area of the shaded region BDC , giving your answer in terms of π and $\sqrt{3}$. [4]

i. Angle $BAC \rightarrow \tan \theta = \frac{2\sqrt{3}}{2}$

$$\theta = \frac{\pi}{3}$$

ii. Area = $\Delta - D$

$$\left(\frac{1}{2} \times 2\sqrt{3} \times 2 \right) - \left(\frac{1}{2} \times 2^2 \times \frac{\pi}{3} \right)$$

$$2\sqrt{3} - \frac{2\pi}{3}$$

SERIES AND PROGRESSION

Formula Based

9709_s22_qp_11

The thirteenth term of an arithmetic progression is 12 and the sum of the first 30 terms is -15 .

Find the sum of the first 50 terms of the progression.

[5]

Solution:

1. $T = a + (n-1)d$
 $a + 12d = 12$
 $a = 12 - 12d$

2. $S_{30} = -15$
 $\frac{30}{2} (2a + 29d) = -15$
 $15 (2(12 - 12d) + 29d) = -15$
 $24 - 24d + 29d = -1$
 $5d = -25$
 $d = -5$
 $a = 12 - 12(-5) = 72$

3. S_{50} :
 $\frac{50}{2} (2(72) + 49(-5))$
 -2525 ANS

SOLUTION:

(i) Find r →

$$\text{Term 4} = ar^3$$

$$64 = (216)r^3$$

$$\sqrt[3]{\frac{64}{216}} = r = \frac{2}{3}$$

(ii) Use formula →

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{216}{1 - \frac{2}{3}} = 648$$

Scenario Based

9709_s20_qp_11

Each year the selling price of a diamond necklace increases by 5% of the price the year before. The selling price of the necklace in the year 2000 was \$36 000.

- (a) Write down an expression for the selling price of the necklace n years later and hence find the selling price in 2008. [3]
- (b) The company that makes the necklace only sells one each year. Find the total amount of money obtained in the ten-year period starting in the year 2000. [2]

Solution:

$$(i) \cdot \text{Increase by 5\%} = 1.05.$$

$$a = 36000$$

$$T = ar^n$$

$$T = 36000(1.05)^n$$

$$T_8 = 36000(1.05)^8$$

$$= \$53200$$

$$(ii) S_{10} = \frac{36000(1.05^{10} - 1)}{1.05 - 1}$$

$$= \$453000$$

DIFFERENTIATION

Equation Of Tangent

9709_s21_qp_11

The equation of a curve is $y = 2\sqrt{3x+4} - x$.

- (a) Find the equation of the normal to the curve at the point (4, 4), giving your answer in the form $y = mx + c$. [5]

$$y = 2\sqrt{3x+4} - x$$

(i) First find gradient by differentiating.

$$\begin{aligned} y &= 2(3x+4)^{1/2} - x \\ &= 2\left(\frac{1}{2}\right)(3x+4)^{-1/2} \times 3 - 1 \end{aligned}$$

$$\frac{dy}{dx} = \frac{3}{\sqrt{3x+4}} - 1$$

(ii). Put (4, 4) to find tangent m then normal.

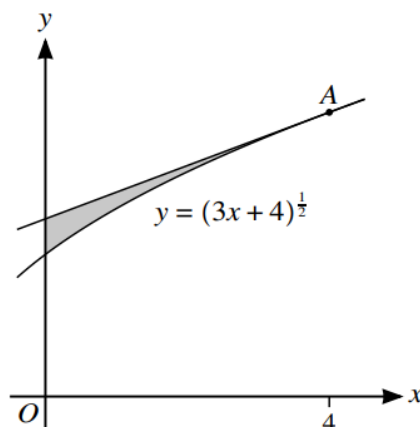
$$m = \frac{3}{\sqrt{3(4)+4}} - 1 = -\frac{1}{4}$$

$$\text{normal} = 4.$$

(iii) make equation using $y - y_1 = m(x - x_1)$

$$y - 4 = 4(x - 4)$$

$$y = 4x - 12$$



The diagram shows part of the curve with equation $y = (3x + 4)^{\frac{1}{2}}$ and the tangent to the curve at the point A. The x-coordinate of A is 4.

- (i) Find the equation of the tangent to the curve at A.

[5]

Solution →

(i) find (x, y)

$$x = 4$$

$$y = (3(4) + 4)^{\frac{1}{2}} = 4.$$

$(4, 4)$

(ii) differentiate:

$$y = (3x + 4)^{\frac{1}{2}}$$

$$= \frac{1}{2} (3x + 4)^{-\frac{1}{2}} \times 3$$

$$= \frac{3}{2} (3x + 4)^{-\frac{1}{2}}$$

$$\text{Put } x=4 \rightarrow \frac{3}{2} (3(4) + 4)^{-\frac{1}{2}} = \frac{3}{8}$$

$$\text{EQ} = y - 4 = \frac{3}{8} (x - 4)$$

$$y = \frac{3}{8}x + \frac{5}{2}$$

dy/dx and Stationary Points

9709_s20_qp_11

The equation of a curve is $y = (3 - 2x)^3 + 24x$.

(a) Find expressions for $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. [4]

(b) Find the coordinates of each of the stationary points on the curve. [3]

i) $y = (3 - 2x)^3 + 24x$	(ii) stationary point. $\frac{dy}{dx} = 0$.	
$\frac{dy}{dx} = 3(3 - 2x)^2(-2) + 24$	$-6(3 - 2x)^2 + 24 = 0$	
$= -6(3 - 2x)^2 + 24$	$(3 - 2x)^2 = 4$	
$\frac{d^2y}{dx^2} = -12(3 - 2x)(-2)$	$3 - 2x = -2$	$3 - 2x = 2$
$= 24(3 - 2x)$	$x = 2.5$	$x = 0.5$
	$y = 64.5$	$y = 20$

9709_s20_qp_12

The equation of a curve is $y = 54x - (2x - 7)^3$.

(a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. [4]

(b) Find the coordinates of each of the stationary points on the curve. [3]

(c) Determine the nature of each of the stationary points. [2]

(i) $y = 54x - (2x - 7)^3$	(ii) $54x - 6(2x - 7)^2 = 0$	
$\rightarrow 54 - 3(2x - 7)^2(2)$	$(2x - 7)^2 = 9$	
$\frac{dy}{dx} = 54 - 6(2x - 7)^2$	$2x - 7 = 3$	$2x - 7 = -3$
$\rightarrow -12(2x - 7)(2)$	$x = 5$	$x = 2$
$\frac{d^2y}{dx^2} = -24(2x - 7)$	(iii) nature:	
	$x = 5$	$-24(2(5) - 7) = -72$ max!
	$x = 2$	$-24(2(2) - 7) = 72$ min!

Rate Change

9709_w22_qp_13

A large industrial water tank is such that, when the depth of the water in the tank is x metres, the volume $V \text{ m}^3$ of water in the tank is given by $V = 243 - \frac{1}{3}(9-x)^3$. Water is being pumped into the tank at a constant rate of 3.6 m^3 per hour.

Find the rate of increase of the depth of the water when the depth is 4m, giving your answer in cm per minute. [5]

$$V = 243 - \frac{1}{3}(9-x)^3$$

$$\frac{dV}{dx} = (9-x)^2$$

$$\frac{dV}{dt} = 3.6$$

$$\frac{dx}{dt} = \frac{1}{(9-x)^2} \times 3.6$$

$$= \frac{1}{(9-4)^2} \times 3.6 = 0.144$$

$$0.144 \text{ m}^3/\text{h} \rightarrow \text{cm}^3/\text{min}$$

$$\frac{0.144 \times 100}{60} = \underline{\underline{0.24}}$$

9709_w21_qp_12.

The volume $V \text{ m}^3$ of a large circular mound of iron ore of radius $r \text{ m}$ is modelled by the equation $V = \frac{3}{2}(r - \frac{1}{2})^3 - 1$ for $r \geq 2$. Iron ore is added to the mound at a constant rate of 1.5 m^3 per second.

- (a) Find the rate at which the radius of the mound is increasing at the instant when the radius is 5.5 m .
[3]

$$\begin{aligned} V &= \frac{3}{2}(r - \frac{1}{2})^3 - 1 \\ \frac{dV}{dr} &\rightarrow \frac{9}{2}(r - \frac{1}{2})^2 \\ r &= 5.5 \\ &= \frac{9}{2}(5.5 - \frac{1}{2})^2 \\ \frac{dV}{dr} &= 112.5 \\ \frac{dr}{dt} &= \frac{dV}{dt} \times \frac{dr}{dV} \\ \frac{dr}{dt} &= 1.5 \times \frac{1}{112.5} \\ &= \frac{1}{75} \text{ m/s.} \end{aligned}$$

INTEGRATION

Equation Of Curve

9709_s22_qp_11

The equation of a curve is such that $\frac{d^2y}{dx^2} = 6x^2 - \frac{4}{x^3}$. The curve has a stationary point at $(-1, \frac{9}{2})$.

Find the equation of the curve.

$\frac{d^2y}{dx^2} \rightarrow \frac{dy}{dx}$	$\frac{dy}{dx} \rightarrow y$
$6x^2 - 4x^{-3}$	$2x^3 + 2x^{-2}$
$\frac{6x^3}{3} - \frac{4x^{-2}}{-2}$	$\frac{2x^4}{4} + \frac{2x^{-1}}{-1}$
$\frac{dy}{dx} = 2x^3 + 2x^{-2} + C$	$y = \frac{1}{2}x^4 - 2x^{-1} + C$
$(0 = \frac{dy}{dx}, x = -1)$	$(-1, 9/2)$
$0 = 2(-1)^3 + 2(-1)^{-2} + C$	$9/2 = \frac{1}{2}(-1)^4 - 2(-1)^{-1} + C$
$C = 0$	$C = 2$
equation: $y = \frac{1}{2}x^4 - \frac{2}{x} + 2$	

9709_s21_qp_11

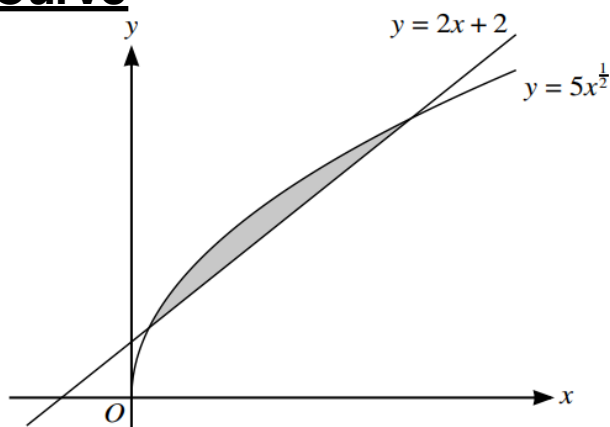
The equation of a curve is such that $\frac{dy}{dx} = \frac{3}{x^4} + 32x^3$. It is given that the curve passes through the point $(\frac{1}{2}, 4)$.

Find the equation of the curve.

[4]

$\frac{dy}{dx} = 3x^{-4} + 32x^3$	$(0.5, 4)$
$\frac{dy}{dx} \rightarrow \frac{3x^{-3}}{-3} + \frac{32x^4}{4}$	$4 = \frac{-1}{(0.5)^3} + 8(0.5)^4 + C$
	$C = 11.5$
$y = \frac{-1}{x^3} + 8x^4 + C$	eq $\rightarrow y = \frac{-1}{x^3} + 8x^4 + \frac{23}{2}$

Area Under Curve



The diagram shows the curve with equation $y = 5x^{\frac{1}{2}}$ and the line with equation $y = 2x + 2$.

Find the exact area of the shaded region which is bounded by the line and the curve.

[5]

AREA:

① first find "x-coordinate" limit by point of intersection

$$y = 2x + 2 \quad y = 5x^{\frac{1}{2}}$$

$$2x + 2 = 5x^{\frac{1}{2}}$$

$$2x - 5x^{\frac{1}{2}} + 2 = 0$$

(solve quadratically)

$$x = 0.25 \quad x = 4.$$

③ Integrate using limits (upper curve) - (lower curve)

$$y = 5x^{\frac{1}{2}}$$

$$y = 2x + 2$$

$$= \left| \frac{5x^{\frac{3}{2}}}{\frac{3}{2}} \right|_{0.25}^4$$

$$= \left| \frac{2x^2}{2} + 2x \right|_{0.25}^4$$

$$\left(\frac{10}{3}(4)^{\frac{3}{2}} \right) - \left(\frac{10}{3}(0.25)^{\frac{3}{2}} \right) - \left((4)^2 + 2(4) \right) - \left((0.25)^2 + 2(0.25) \right)$$

$$\frac{105}{4}$$

$$\frac{375}{16}$$

$$= \frac{45}{16}$$

Q2)

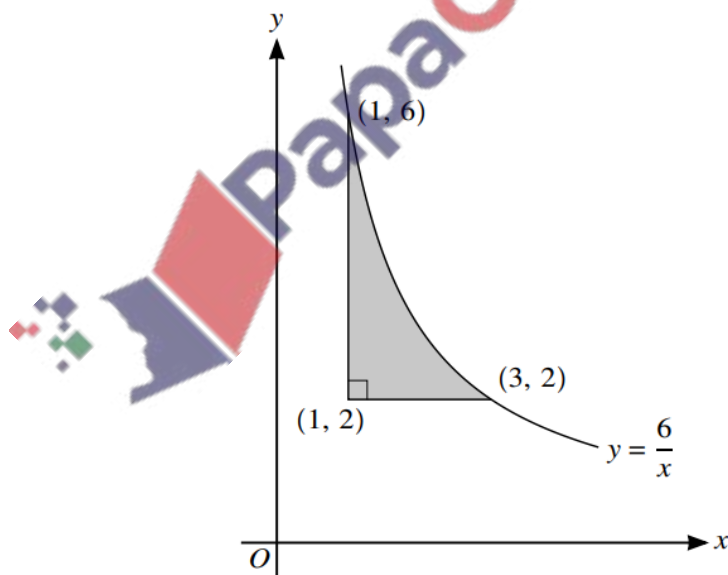
Find the area of the region enclosed by the curve $y = 2\sqrt{x}$, the x -axis and the lines $x = 1$ and $x = 4$.

[4]

$$\begin{aligned} (1) \text{ integrate} \\ y &= \int 2x^{1/2} \\ &= \frac{2x^{3/2}}{3/2} \\ (2) \text{ limit: } &\left| \frac{4}{3} x^{3/2} \right|_1^4 \\ &= \frac{4}{3} (4)^{3/2} - \frac{4}{3} (1)^{3/2} = \frac{28}{3} \end{aligned}$$

Volume Under Curve

9709_s20_qp_12



The diagram shows part of the curve $y = \frac{6}{x}$. The points $(1, 6)$ and $(3, 2)$ lie on the curve. The shaded region is bounded by the curve and the lines $y = 2$ and $x = 1$.

(a) Find the volume generated when the shaded region is rotated through 360° about the **y-axis**. [5]

"about y-axis" $\rightarrow$ make x subject.

$$x = \frac{6}{y}$$

UPPER CURVE

$$x^2 = 36y^{-2}$$

$$\text{integrate: } \int_2^6 36y^{-2}$$

$$\left| -36y^{-1} \right|_2^6$$

$$\left| \left(-36(6)^{-1} \right) - \left(-36(2)^{-1} \right) \right|$$
$$\left| (-6) - (-18) \right|$$

$$= 12 \times \pi = 12\pi$$

CYLINDER

$$= \pi r^2 h$$

$$= \pi (1)^2 4$$

$$= 4\pi$$

$$\text{FINAL} = 12\pi - 4\pi$$

$$= 8\pi$$

TRIGONOMETRY

Prove The Identity

9709_s20_qp_11

(a) Prove the identity $\frac{1 + \sin \theta}{\cos \theta} + \frac{\cos \theta}{1 + \sin \theta} \equiv \frac{2}{\cos \theta}$.

[3]

Solution:

$$(1) \text{ LCM: } \frac{(1 + \sin \theta)(1 + \sin \theta) + (\cos \theta)(\cos \theta)}{\cos \theta (1 + \sin \theta)}$$

$$\Rightarrow \frac{1 + \sin^2 \theta + \sin \theta + \sin \theta + \cos^2 \theta}{\cos \theta (1 + \sin \theta)}$$

$$\Rightarrow \frac{1 + 1 + 2 \sin \theta}{\cos \theta (1 + \sin \theta)}$$

$$\Rightarrow \frac{2(1 + \sin \theta)}{\cos \theta (1 + \sin \theta)}$$

$$\Rightarrow \frac{2}{\cos \theta}$$

9709_w20_qp_11

(a) Show that $\frac{\sin \theta}{1 - \sin \theta} - \frac{\sin \theta}{1 + \sin \theta} \equiv 2 \tan^2 \theta$.

[3]

(1) LCM:

$$\frac{\sin \theta (1 + \sin \theta) - \sin \theta (1 - \sin \theta)}{1 - \sin^2 \theta}$$

$$\frac{\sin^2 \theta + \sin \theta - \sin \theta + \sin^2 \theta}{1 - \sin^2 \theta}$$

$$\frac{2 \sin^2 \theta}{\cos^2 \theta} = 2 \tan^2 \theta$$

Find The Angle

9709_w22_qp_13

Solve the equation $8 \sin^2 \theta + 6 \cos \theta + 1 = 0$ for $0^\circ < \theta < 180^\circ$.

[3]

SOLUTION

$$8(1 - \cos^2 \theta) + 6 \cos \theta + 1 = 0$$

$$8 - 8 \cos^2 \theta + 6 \cos \theta + 1 = 0$$

$$8c^2 - 6c - 9 = 0$$

$$\cos \theta = 3/2$$

$$\cos \theta = -3/4$$

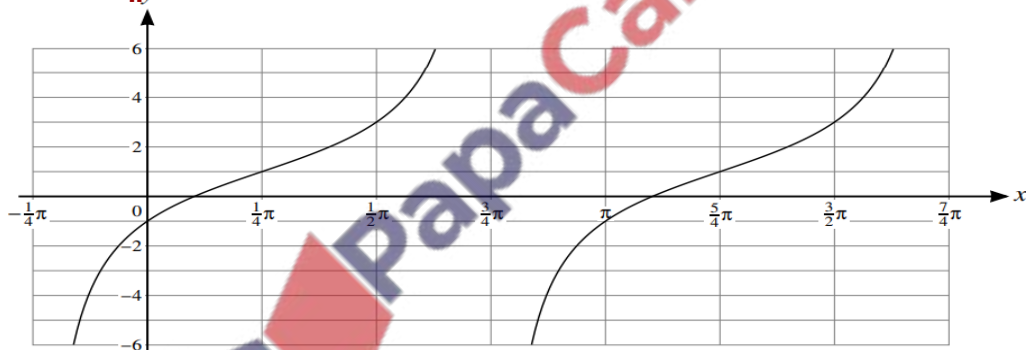
$$\theta = \text{not pos.}$$

$$\theta = 41.41$$

$$\text{ans} = 138.6^\circ \text{ (neg quadrant)}$$

Graph

9709_s21_qp_11



The diagram shows part of the graph of $y = a \tan(x - b) + c$.

Given that $0 < b < \pi$, state the values of the constants a , b and c .

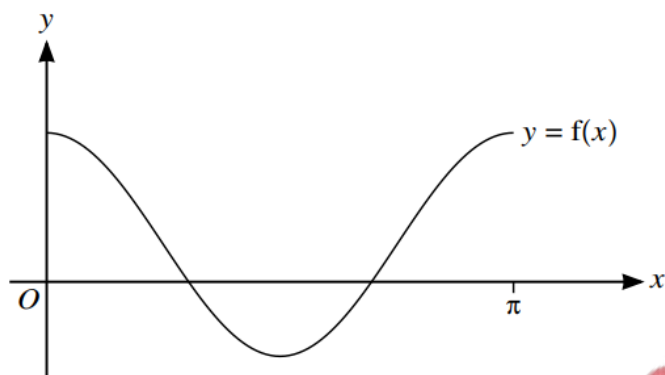
[3]

SOLUTION:

$$a = \text{amplitude} = 2$$

$$b = \text{transformed right by } \pi/4.$$

$$c = 1 / \text{constant line.}$$



The diagram shows the graph of $y = f(x)$, where $f(x) = \frac{3}{2} \cos 2x + \frac{1}{2}$ for $0 \leq x \leq \pi$.

(a) State the range of f .

[2]

To find range of f :

sub / add amplitude to constant line.

$$\frac{3 \cos 2x}{2} + \frac{1}{2} \quad \bullet \quad \frac{1}{2} + \frac{3}{2} = 2$$

$$\frac{3 \cos 2x}{2} + \frac{1}{2} \quad \bullet \quad \frac{1}{2} - \frac{3}{2} = -1$$

amplitude constant line = $-1 \leq f(x) \leq 2$

FUNCTIONS:

With Differentiation

9709_w21_qp_13

- (b) The function f is defined by $f(x) = x^5 - 10x^3 + 50x$ for $x \in \mathbb{R}$.

Determine whether f is an increasing function, a decreasing function or neither.

[3]

Solution: differentiate first.

$$f(x) = x^5 - 10x^3 + 50x$$

$$\frac{dy}{dx} = 5x^4 - 30x^2 + 50$$

factorise now:-

$$5x^2(x^2 - 6 + 3^2) - (3)^2 \times 5 + 50$$

$$5x^2(x^2 - 3)^2 + 5$$

$\therefore$ Hence, as, for any value of x

$$(x^2 - 3)^2 \text{ will be } > 0$$

$$5(x^2 - 3)^2 \text{ will be } > 0$$

$$5(x^2 - 3)^2 + 5 \text{ will be } > 0$$

as always positive! It is increasing function

Composite Functions

9709_w21_qp_12.

The function f is defined as follows:

$$f(x) = \frac{x+3}{x-1} \text{ for } x > 1.$$

- (a) Find the value of $ff(5)$.

[2]

$$f(x) = \frac{x+3}{x-1}$$

input 5 input $f(5)$ in x

$$f(5) = \frac{5+3}{5-1} = 2 \quad \bullet \quad ff(5) = \frac{2+3}{2-1} = 5$$

Transformations

9709_w21_qp_12.

The graph of $y = f(x)$ is transformed to the graph of $y = f(2x) - 3$.

- (a) Describe fully the two single transformations that have been combined to give the resulting transformation. [3]

$$y = f(2x) - 3$$

1: stretch by factor of $\frac{1}{2}$ in x -direction 2: translate by $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$

9709_w21_qp_13

The graph of $y = f(x)$ is transformed to the graph of $y = 3 - f(x)$.

Describe fully, in the correct order, the two transformations that have been combined.

[4]

$$y = 3 - f(x)$$

1: reflection in x -axis
2: translate by $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$

BINOMIAL THEOREM

Expand First few Terms

Q1) 9709_w21_qp_13

- (a) Find the first three terms, in ascending powers of x , in the expansion of $(1 + ax)^6$.

$$\begin{array}{l} (1 + ax)^6 \\ \hline {}^6C_0 ax^0 |^6 + {}^6C_1 ax^1 |^5 + {}^6C_2 ax^2 |^4 \\ \hline 1 + 6ax + 15a^2x^2 \end{array}$$

Q2) 9709_w22_qp_13

- (a) Find the first three terms in ascending powers of x of the expansion of $(1 + 2x)^5$.

$$\begin{array}{l} (1 + 2x)^5 \quad {}^nC_r(b^r)(a^{n-r}) \\ \hline {}^5C_0 2x^0 |^5 + {}^5C_1 2x^1 |^4 + {}^5C_2 2x^2 |^3 \dots \\ \hline 1 + 10x + 40x^2 \end{array}$$

FIND COEFFICIENT OF SPECIFIC TERMS

9709_s21_qp_11

(b) Hence find the coefficient of x^2 in the expansion of $(4+x)^2(3-2x)^5$.

[3]

coeff of x^2 :

$$(4+x)^2(3-2x)^5$$

$$(16+8x+x^2)({}^5C_0 2x^0 3^5 - {}^5C_1 2x^1 3^4 + {}^5C_2 2x^2 3^3 \dots)$$

$$(16+8x+x^2)(243 - 810x + 1080x^2 \dots)$$

$$243x^2 - 6480x^2 + 17280x^2$$

$$11043x^2$$

$$= 11043$$

9709_w21_qp_12.

(a) It is given that in the expansion of $(4+2x)(2-ax)^5$, the coefficient of x^2 is -15 .

Find the possible values of a .

[4]

coeff of $x^2 = -15$

$$(4+2x)(2-ax)^5$$

$$(4+2x)({}^5C_0 ax^0 2^5 - {}^5C_1 ax^1 2^4 + {}^5C_2 ax^2 2^3 \dots)$$

$$(4+2x)(32 - 80ax + 80a^2x^2 \dots)$$

$$-160ax^2 + 320a^2x^2 = -15x^2$$

$$320a^2 - 160a + 15 = 0$$

$$a = 1/8 \quad \text{or} \quad a = 3/8$$